

Systematic Supremacy

Writing in 1923 about the famous discretionary speculator Jesse Livermore, the American author Edwin Lefèvre captured in a quote the frailties of human psychology when it comes to investing in financial markets:

“It is inseparable from human nature to hope and to fear. In speculation when the market goes against you – you hope that every day will be the last day – and you lose more than you should had you not listened to hope... And when the market goes your way you become fearful that the next day will take away your profit, and you get out – too soon. Fear keeps you from making as much money as you ought to.”

In this letter, I am explaining the key advantages of the systematic approach to investing. In addition to the avoidance of investment error due to psychological bias, a systematic approach offers several key benefits including: the scalability to invest with a consistent approach twenty-four hours per day across a global portfolio of securities; the implementation of consistent risk management at security, asset class and portfolio level; and the scientific rigor which can be devoted to the continuous development of the core investing approach.

1. Human Psychology

Up until the dotcom bubble academic finance was slow to accept the link between investor psychology and financial market behavior. However, the field of behavioral finance is now well established, identifying two main areas of relevance to a discussion of systematic investing. First, there is now a considerable body of theoretical and empirical evidence identifying how difficult it is for discretionary investors to make consistently reliable forecasts and decisions about risk allocation. Second, there is now recognition of the effect these psychological biases have on financial markets and the consequent opportunities for systematic investors.

1.1. Discretionary trading and human psychology

In 1979 two psychologists, Daniel Kahneman and Amos Tversky, published a groundbreaking paper on decision making under uncertainty. The authors describe a theory supported by experimental evidence, which predicts (irrational) attitudes to risk aversion and risk seeking, when individuals are making decisions with no certainty about the outcome.

The principle decision making error by individuals is that they avoid crystallizing small losses (risk seeking when it comes to small losses) while at the same time realize profits too soon (risk averse when it comes to profits). Though the focus of the two psychologists’ work was not financial markets, this is a feature of investing behavior that was recognized at least as far back as Jesse Livermore, in the quote above.

Interestingly, there is also a further element of Kahneman and Tversky’s theory which gets less attention. People are risk seeking when it comes to the opportunity to make very large profits (the large demand for lotteries being a classic example) and generally avoid the small probability of very large losses (evidenced by the large demand for insurance products).

In related research Kahneman, Tversky and others have documented several additional errors that individuals tend to make around decisions, due to psychological biases. For example, generally individuals will rely on relatively little historical information to make predictions with high confidence (the law of small numbers), will tend to be overconfident about their own forecasting abilities and are generally overly optimistic. This is particularly acute when it comes to estimating the probability of future events occurring.

Subsequently, scientific literature has evolved to incorporate a deeper understanding of how financial market participant biases affect decision making. These biases are driven by additional factors which include cultural issues and extreme life events, some of which are documented in Fig.1.

There is strong evidence that individual investors like to buy stocks with lottery-like characteristics. This phenomenon is particularly prevalent in areas where there is high lottery participation and appears to be driven by cultural issues. Factors which induce greater expenditure in lotteries also induce greater investment in lottery-type stocks. Poor, young men who live in urban, Republican dominated regions and belong to specific minority and religious groups invest more in lottery type stocks.

Extreme life events such as marriage and divorce have been shown to affect fund manager performance. In the years surrounding both marriage and divorce investors exhibit lower realized returns, their stock selection skills are poorer, and their risk adjusted returns deteriorate, and are weaker compared to control samples. This also holds for hedge fund managers.

Recent evidence by researchers at the University of Miami documents investors tending to favor funds managed by managers from a similar cultural background to themselves. Their evidence shows that US mutual funds managed by individuals with foreign sounding names experience 10% per annum lower inflows than otherwise identical funds.

All of the above evidence points to the human frailties which make discretionary investing so difficult to pursue successfully over a long period in a consistent fashion.

1.2. How human behavior generates opportunities for systematic traders

There is also growing scientific evidence demonstrating how the human weaknesses discussed in the preceding section create opportunities in financial markets which can be systematically exploited. The most recognized example is how behavioral models link what we know about investor overconfidence and changes in risk aversion to initial underreaction and then over-reaction to new information and consequently cause momentum (trends) in asset prices.

By maintaining the discipline of operating systematically, an adaptable systematic strategy manages both to avoid human behavioral biases, and to exploit the trends created by this behavior. This example emphasized how effective the simple philosophy of running profits and cutting losing positions can be, but how difficult it is in practice for discretionary investors to achieve.

The graph illustrates the following market behaviors and investor sentiments:

- Initial Point:** (buy) - Price is low.
- First Peak:** Anchoring (do nothing) - "I'm not paying attention to the news. It's worth more than this."
- First Trough:** Disposition (sell) - "At least I didn't sell at the low!"
- Second Peak:** Herding (buy) - "I'm missing out."
- Second Trough:** Disposition (do nothing) - "I will sell when things look better."
- Third Trough:** Overreaction (sell) - "I'm out!"
- Final Peak:** Herding (buy) - "I'm in ... again!"

A systematic investment approach offers key additional benefits in terms of implementation, particularly when operated at scale. Systematic managers such as SINTRO are able to trade globally and operate around the clock. A typical system in the futures space can trade well over 100 assets on multiple exchanges around the world, and a systematic equity trading strategy may monitor and trade in several thousand stocks. Because all of the decision making and investment processes are automated, processing of new information, risk monitoring, signal generation and trade execution can be implemented continuously.

This scalability of both markets traded and strategies which can be utilised increases the potential diversification in a systematically operated portfolio. And this in turn means that more potential trading strategies become worthwhile pursuing: systematic traders can focus on marginal effects in the markets which may only have a small, expected benefit. A strategy with a small edge might not be attractive or practical to operate in a manual or discretionary manner but might yield far more compelling results when spread over many markets and traded repeatedly.

The implementation of a systematic investment approach also allows for major innovation in disciplined risk management, which can be built into the strategy rather than applied as an afterthought. Typically, a systematic manager will operate a targeted level of portfolio risk, while some managers will allow the risk level to vary within a target range.

Statistical techniques allow for relatively accurate forecasting of short-term volatility. This means systematic managers are able to ensure realized asset level risk lies within a relatively narrow band.

Augmenting this with correlation forecasts allows the portfolio to be constructed using rigorous statistical methods.

Typically, a systematic manager will begin by generating risk forecasts for the different markets in the portfolio and estimating the correlation between different assets. The portfolio is constructed by combining these estimates and the signals generated by the trading models.

In addition to relying on scientific methods to target volatility, the systematic approach allows hard limits to be set and monitored to control risks and exposures. For example, these could include hard limits for volatility, Value at Risk and leverage at the asset, asset class and portfolio level. Once these limits are reached position sizes can be automatically capped or reduced.

4. Scientific Approach

SINTRO's investment philosophy has remained consistent since the company's inception and focuses on a scientific approach to investment driven by our belief that market prices are not random but display persistent, statistically measurable, and predictable behavior and idiosyncrasies.

SINTRO has an extremely high-quality proprietary and algorithmic infrastructure to handle and execute depending on real-time financial data. Initial testing and development are conducted on a sub-sample of the historical dataset. If the hypothesis is not rejected in the initial test, the test is repeated using the markets and historic periods which had been kept out of sample. This multi-stage approach minimizes data mining pitfalls. The next step in the research process is to examine the sensitivity of results to assumptions. Alternative methods of testing the hypothesis are also specified by an independent risk review which validates all research findings and ensures that each new strategy fits well into the overall portfolio. Finally, it is possible to monitor the prototype strategy in a paper trading period to assess real world performance, prior to commissioning. This rigorous scientific method is in stark contrast to the rather ad-hoc approach of a typical discretionary manager.

6. Conclusion

Investors who have embraced systematic techniques more recently were perhaps initially influenced by their very strong performance in the depths of the global financial crisis. But in our regular meetings with clients, we have noticed tremendous advances in awareness of the broader benefits of a systematic approach in all market environments.

This bodes well for the future of the investment industry. By harnessing the advantages of a scientific approach to investment management outlined in this paper, clients should be well placed to face the uncertainty inherent in investing in financial markets.